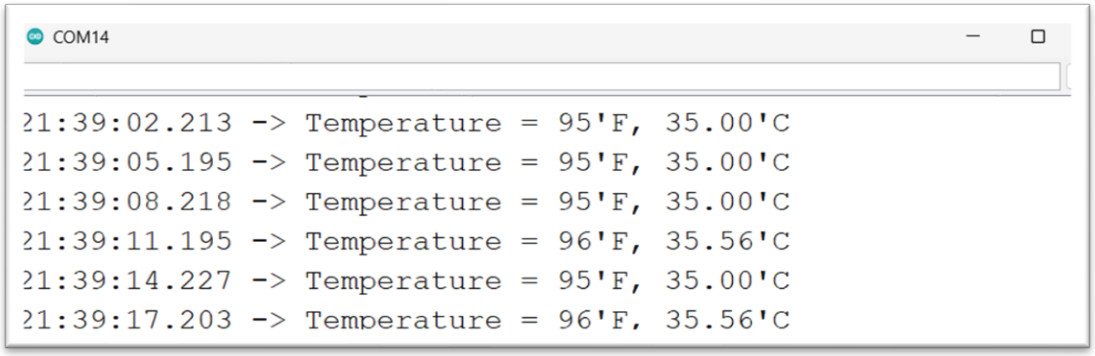
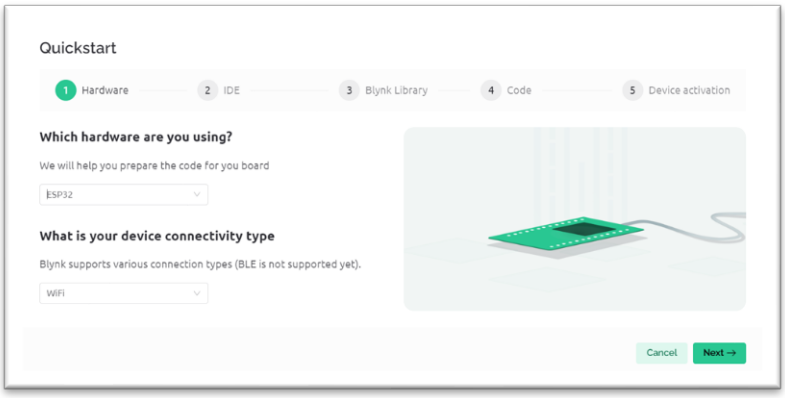
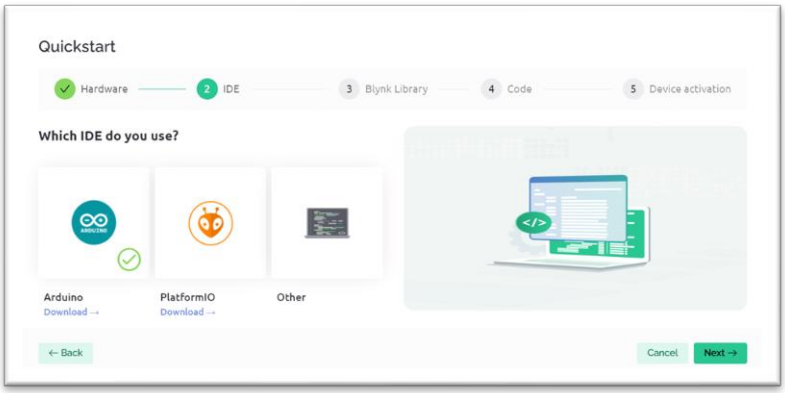
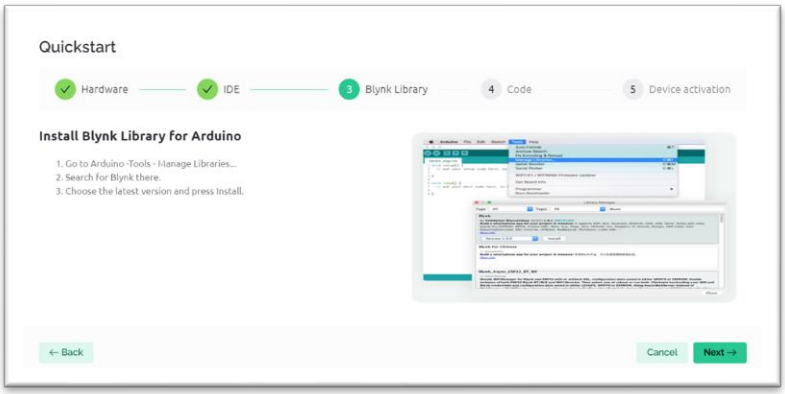
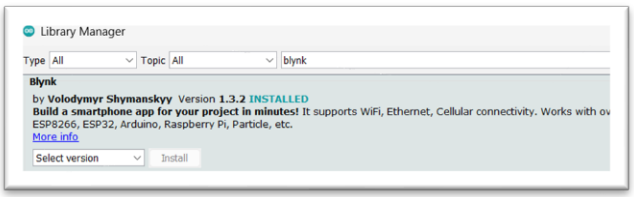
0/3 -- ESP32 Internal Temperature Sensor

```
// built in temperature sensor
// http://fitrox.lnwshop.com/article/45/esp32-ตอนที่-3-esp32-built-in-temperature-sensor

extern "C" {
  uint8_t temprature_sens_read();
}

void setup() {
  Serial.begin(115200);
}

void loop() {
  int fTempp = temprature_sens_read();
  float cTempp = (fTempp - 32) / 1.8;
  Serial.print("Temperature = ");
  Serial.print(fTempp);
  Serial.print("'F, ");
  Serial.print(cTempp);
  Serial.println("'C");
  delay(3000);
}
```



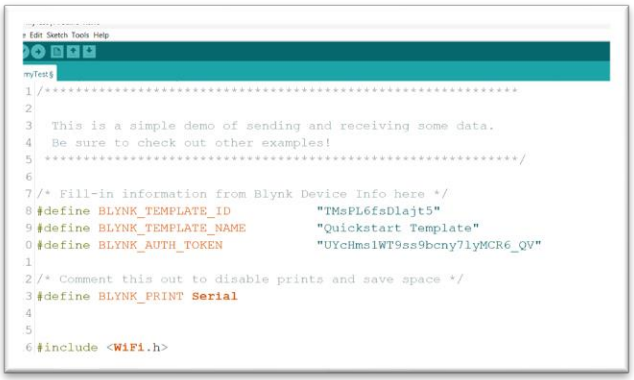
1/3 – Blynk 2.0

- Sign in ➔ Token ID
- Create Template ➔ ESP32, WiFi
- Create Device on Template ➔ Device Name, Device ID
- Crete DataStream on Template

Exp11: Hello Blynk	
• https://blynk.io/	
• ESP32 • WiFi	
• Arduino IDE	
• Add Library	
Blynk by Volodymyr Shymanskyy Version 1.3.2	

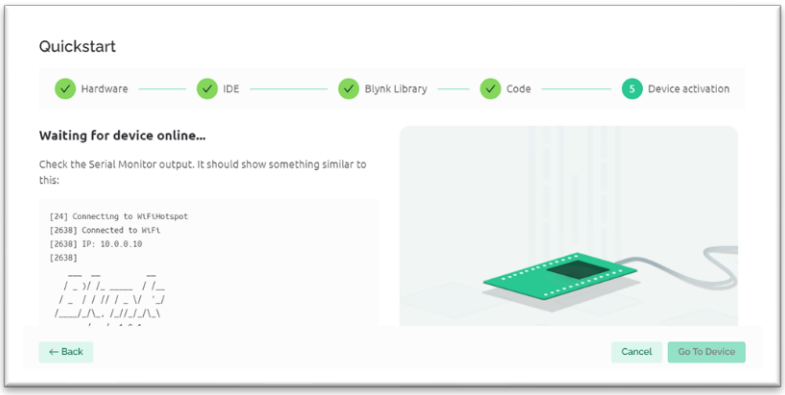
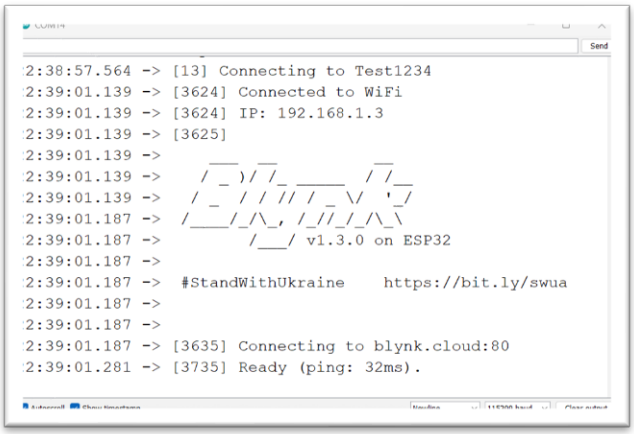
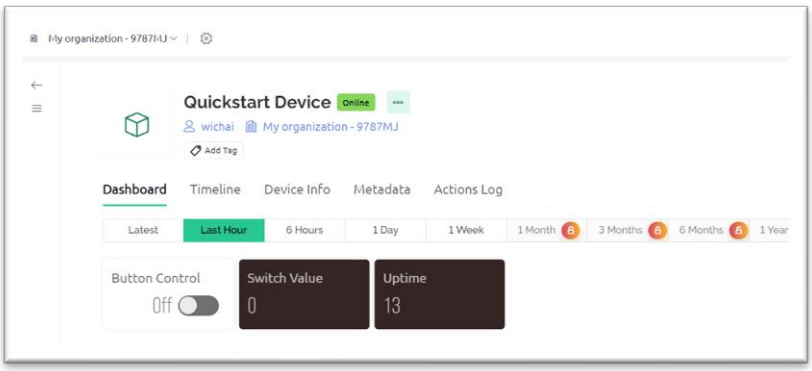
- Wifi ID, Pass
- Create Code

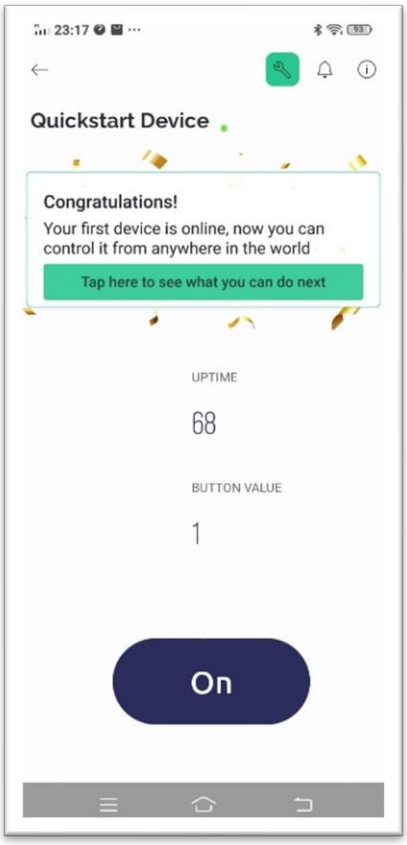
- Copy to Arduino IDE



- **Keep: ID, Name, Token**
- **KEEP: SSID, Pass**
- **Chang All Code to …**

```
/* Fill-in information from Blynk Device Info here */
#define BLYNK_TEMPLATE_ID        "TMPL5Dlajt5"
#define BLYNK_TEMPLATE_NAME   "Quickstart Template"
#define BLYNK_AUTH_TOKEN         "UYkchcHm1WT1ny7lyMCRA6_QV"

#define BLYNK_PRINT Serial
#define LED_Test 2

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>

char ssid[] = "Test134";
char pass[] = "081799";

BlynkTimer timer;

BLYNK_WRITE(V0) {
  // Set incoming value from pin V0 to a variable
  int value = param.asInt();
  // Update state
  Blynk.virtualWrite(V1, value);
  digitalWrite(LED_Test, value);
}

void myTimerEvent() {
  Blynk.virtualWrite(V2, millis() / 1000);
}

void setup() {
  Serial.begin(115200);
  pinMode(LED_Test, OUTPUT);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timer.setInterval(1000L, myTimerEvent);
}

void loop() {
  Blynk.run();
  timer.run();
}
```

• Load to Board	
• ESP32 Run and Connected Blynk Server	
• Goto Blynk Web • Control LED D2	

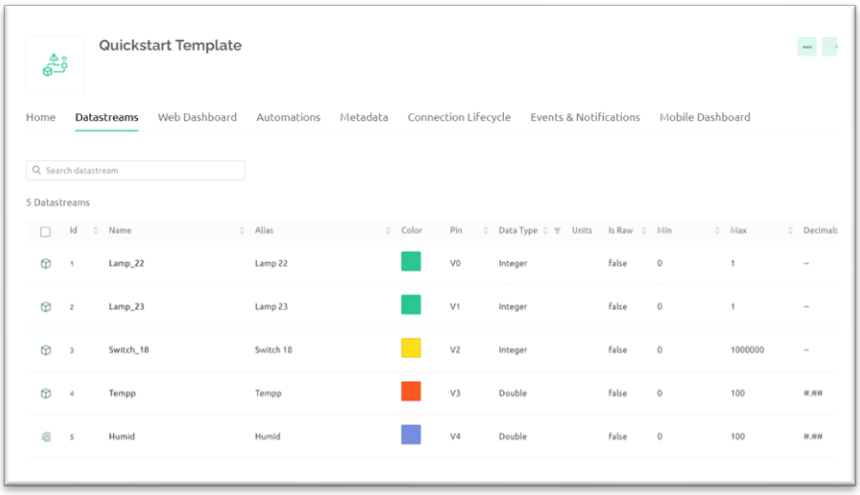
• Install Blynk App • Open Blynk • Control LED D2	

Exp12: Blynk 2 LED + 1Sw + ~~DHT22~~ Random

- Blynk.io
- Edit Dashboard
- Data Stream

Lamp22, V0, Integer [0-1]
Lamp23, V1, Integer [0-1]
Switch18, V2, Integer [0-1]
Tempp, V3, Double [0-100] #.##
Humid, V4, Double [0--100] #.##

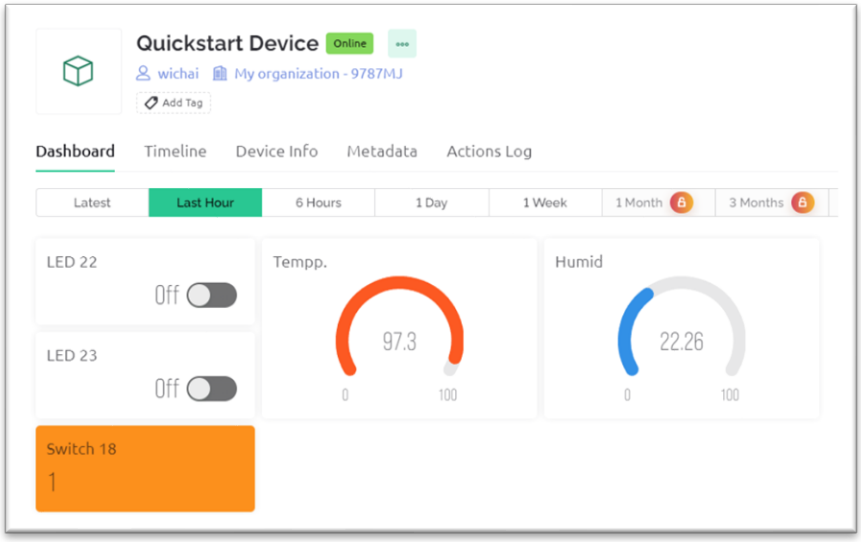
- Save and Apply



- Edit Dashboard

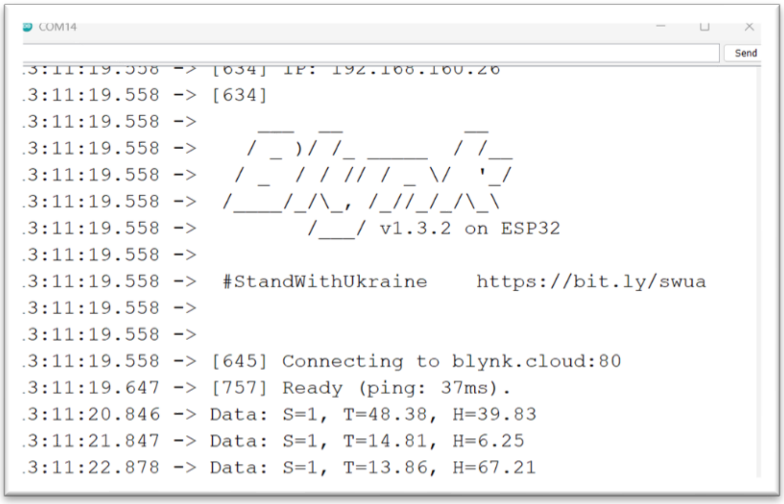
LED_22 = V0
LED_23 = V1
Switch_18 = V3
Tempp. = V4
Humid. = V5

- Save and Apply

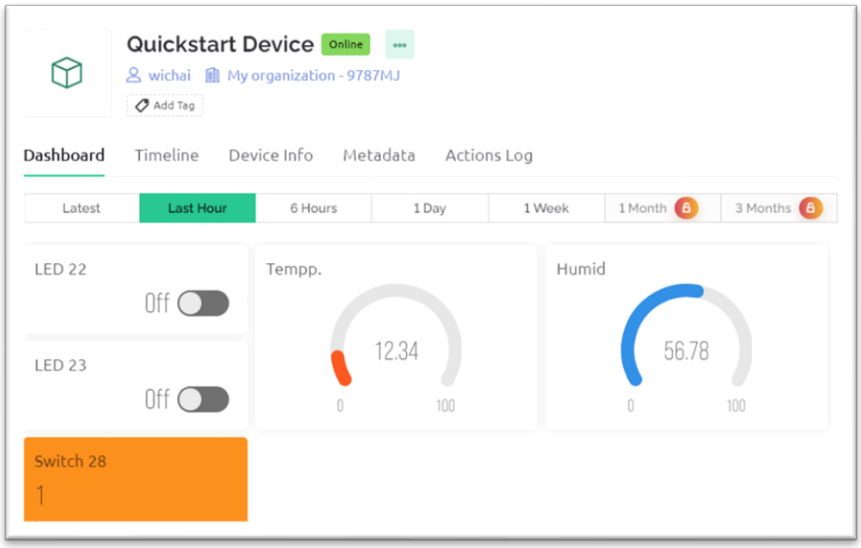


- Code and Load
- **Keep: ID, Name, Token**
- **KEEP: SSID, Pass**

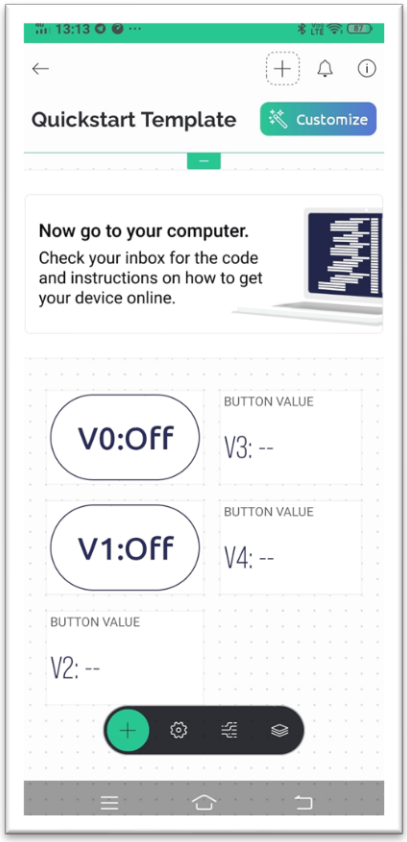
```
/* Fill-in information from Blynk Device Info here */
#define BLYNK_TEMPLATE_ID         "TML6f5Dlaj5"
#define BLYNK_TEMPLATE_NAM     "Quickstart Template"
#define BLYNK_AUTH_TOKEN          "UYkchc1W9s9bcyLny7lyMCRA6_QV"

#define BLYNK_PRINT Serial
#define LED_22    22
#define LED_23    23
#define SWitch_18 18

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>

char ssid[] = "T-Test";
char pass[] = "pk345678";

BlynkTimer timer;
char sBuffer[60];

BLYNK_WRITE(V0) {
  int value = param.asInt();
  digitalWrite(LED_22, value);
}

BLYNK_WRITE(V1) {
  int value = param.asInt();
  digitalWrite(LED_23, value);
}

void myTimerEvent() {
  int value = digitalRead(SWitch_18);
  float Tempp = random(9999) / 100.0;
  float Humid = random(9999) / 100.0;
  Blynk.virtualWrite(V2, value);
  Blynk.virtualWrite(V3, Tempp);
  Blynk.virtualWrite(V4, Humid);
  sprintf(sBuffer, "Data: S=%d, T=%0.2f, H=%0.2f", value, Tempp, Humid);
  Serial.println(sBuffer);
}

void setup() {
  Serial.begin(115200);
  pinMode(LED_22, OUTPUT);
  pinMode(LED_23, OUTPUT);
  pinMode(SWitch_18, INPUT_PULLUP);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timer.setInterval(1000L, myTimerEvent);
}

void loop() {
  Blynk.run();
  timer.run();
}
```

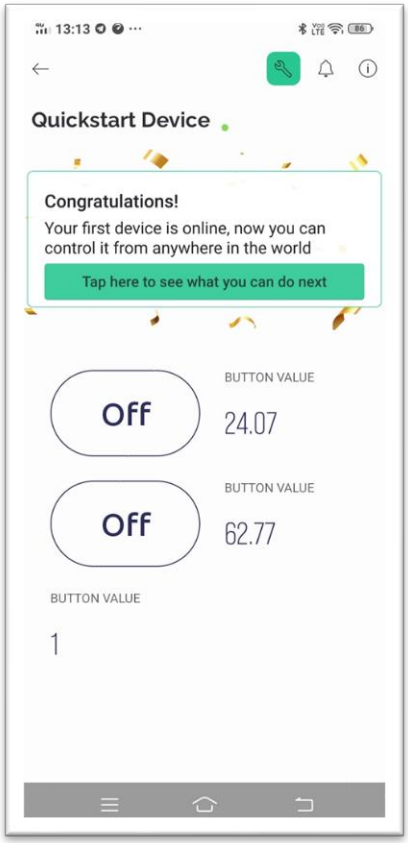


- Run / Control



- Edit Mobile App
- And Run





Exp13: Blynk 2 LED + 1Sw + DHT22 + Display on TM1638 (or MAX7219)
• ปรับจาก random ให้อ่าน DHT22 • เพิ่มเติมให้แสดง Tempp และ Humid ที่หน่วยแสดงผล

2/3 – IFTTT to Google Sheet and LINE

3/3 – Direct to Google Sheet